Maxwell Fang

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EDUCATION

BMath Honours Mathematics Ontario, Canada
University of Waterloo Anticipated April 2026

- Major in Combinatorics and Optimization
- Formerly enrolled in Honours Mathematics Co-op

RESEARCH EXPERIENCE

Undergraduate Research Assistant
University of Waterloo, Waterloo, ON Nov 2024 – Present

- *Supervisor*: Dr. Stephen Vavasis
- Primary Collaborator (2-person team)
- Validated existence of neural scaling laws in natural image, audio, and text datasets
- Derived and tested closed-form solutions and approximations for eigenvalues of covariance matrices of artificial datasets
- Developed a method to generate artificial image datasets modeling neural scaling laws using numpy and OpenCV to analyze eigenpairs of the covariance matrix
- Conducted numerical experiments on MNIST and Artificial datasets in Matlab to understand and determine causes of diagonalization and eigenvector localization in Fourier transformed covariance matrices of artificial datasets

PROFESSIONAL EXPERIENCE

Software Developer – AI/ML
Ford Motor Company of Canada, Waterloo, ON Jan 2025 – April 2025

- Designed and implemented Perspective View to Bird's Eye View image transformation module and inference visualization using LIDAR and RADAR ground truths with Python, cv2, and numpy
- Developed image interpolation to correct visual artifacts in post-transformation Bird's Eye View images and motion smoothing to augment low framerate videos
- Created visualization pipeline using PyGObject Gstreamer and threading to evaluate accuracy of LIDAR/RADAR ground truths and lane-line prediction model by visualizing inferences on PV images and converting to motion-smoothed video

Lead AI/ML Engineer Co-op
Eon Media, Toronto, ON May 2024 – August 2024

- Led and trained a team of 3 coops to maintain pre-existing video processing pipelines, object detection models, NLP models, and develop AI/ML solutions
- Implemented Robust Continuous Clustering Dimensionality Reduction with Conjugate Gradient Method and PCA to cluster high dimensional image embeddings (2048 dimensions) created by Pytorch resnet50 and reduce former clustering runtime 75% using numpy, scipy.sparse and concurrent.futures
- Designed and implemented Logo detection model training pipeline to reduce manual labour by 45% and 3rd-party software costs by 80%, featuring image clustering, data cleaning, formatting, and augmentation for YOLO-based models

AI/ML Engineer Co-op
Eon Media, Toronto, ON January 2024 – April 2024

- Optimized and redesigned video processing pipelines using Python multiprocessing and Kubernetes to decrease CPU usage by 85%, pipeline setup time by 60% and overall processing time by 40%
- Designed and minimized runtime of pre-existing object detection models to implement a Reverse Image Search that identifies brands, people and objects in images and returns keywords in < 400 ms hosted on Flask with Nginx, and Gunicorn
- Created and Containerized python and CUDA based applications for metadata extraction from video (brands, people, location, context, etc.) using Docker

QA Automation Intern
CAA South Central Ontario, Thornhill, ON May 2022 - August 2022

- Produced reusable automation modules in Ranorex and Eggplant functional to conduct data-driven and functional testing in Guidewire and SAP business client
- Created and maintained testing data and output data in MS SQL Server Management studio and increased test-case automation by 200%
- Optimized test scripts by redesigning data-importation modules to improve flexibility in existing modules and more efficiently import testing data from SQL Server

VOLUNTEER EXPERIENCE

Deputy Vice President, Academics
Undergraduate Mathematics Society of the University of Waterloo, Waterloo, ON Sept 2025 – Present

- Canvassed Mathematics students on campus to gather sentiment and feedback on proposed changes to courses, plans, programs, and departments
- Planned and organized Resume Review events for 50+ students and offered feedback and advice for coop placements and resume improvements

4th Year Undergraduate/CO Representative
Math Endowment Fund, University of Waterloo, Waterloo, ON May 2025 – Present

- Allocated ~$100, 000 of funding for clubs, organizations, and events benefiting the Mathematics student body through the funding of extracurriculars, design teams, research programs, and networking events
- Discussed and reviewed project funding proposals with other student representatives to determine funding quantities

SCHOLARSHIPS AND AWARDS

President's Scholarship of Distinction, University of Waterloo 2021

SKILLS

- Programming Languages: Python, Matlab, SQL, R, Bash
- Tools: Git, Docker, Kubernetes, AWS EC2, EKS, ECR, S3, Gstreamer, Microsoft Office, Tableau, LaTex
- RDBMS: MS SQL Server Management studio, SQL Developer, Microsoft SQL server 19

LANGUAGE SKILLS

- Native/Advanced: English
- Intermediate: French
- Basic: Mandarin